Assignment #1 – Computer Networks (Fall 2025)

Q1. Single packet across two links with a router

(a) Total end-to-end delay for sending a single packet of length L across two links with rates R1 and R2 and propagation delays d1 and d2 is:
D = (L/R1) + d1 + (L/R2) + d2 (assuming negligible processing delay).
Explanation: (L/R1) = transmission delay on link 1, d1 = propagation delay on link 1, (L/R2) = transmission delay on link 2, d2 = propagation delay on link 2. Router queueing delay may also appear if traffic > capacity.
(b) If R2 << R1, then transmission on link 2 dominates the delay. The router becomes a bottleneck, packets may queue, and the effective end-to-end delay is approximately (L/R2) + d1 + d2.

Q2. HFC vs DSL

(a) DSL provides a dedicated twisted pair copper line to the DSLAM, so local loop bandwidth is not shared. HFC uses a coaxial segment shared by many neighbors; bandwidth is divided among active users.
(b) HFC users experience slowdown at 8:00 PM because many neighbors share the same coaxial bandwidth simultaneously. DSL users don't suffer the same neighborhood contention since their local loop is dedicated.

Q3. Web page request headers

(a) Headers are added at multiple layers:
1. Application layer (HTTP): request line, Host, User-Agent. Role: tells server what resource is requested.
2. Transport layer (TCP): src port, dst port, sequence numbers. Role: multiplexing, reliable delivery.
3. Network layer (IP): src IP, dst IP, TTL. Role: routing between hosts.
4. Link layer (Ethernet): src MAC, dst MAC. Role: delivery on local link.

(b) Routers examine only the network (IP) header (specifically destination IP and TTL). They ignore transport and application headers for forwarding.
(c) Segmentation advantage: allows pipelining and retransmission of only lost parts. Disadvantage: header overhead and more processing per segment.

Q4. Users on a 1 Gbps link

(a) Circuit switching: each user requires 100 Mbps. 1 Gbps/100 Mbps = 10 users max.
(b) Packet switching: more than 10 users possible since not all are active simultaneously. Statistical multiplexing makes better use of link capacity.

Q5. Asking directions analogy

(a) Car = packet; Driver = process/routing logic; Intersections = routers; Destination = host application.
(b) A fake detour sign corresponds to a false route advertisement (e.g., BGP hijack).
(c) Security principle violated: Integrity of routing information.

Q6. End-of-chapter problems

Hints for problems:
1. P5: use L/R + distance/speed formulas for transmission and propagation delay.
2. P6: compute bandwidth-delay product = bandwidth × RTT, represents bits in flight.
3. P18: queuing delay grows quickly as utilization approaches 1.
4. P24: compare statistical multiplexing vs circuit switching, often using probability that k users active.
5. P31: analyze segmentation/ACK timing, compute using RTT and segment size.

Bonus: Ayesha in Karachi

(a) Single packet one-way delay = 41.85 ms. Computed by summing per-hop transmission (L/R), propagation (distance/speed), and queueing (given). Submarine hop dominates.

(b) 10 GB file = 80×10^9 bits. Bottleneck link = 1 Gbps. Transfer time = 80s plus ≈ 0.08s handshake/RTT. Total ≈ 80.08s.

End of Detailed Solutions